

RL Bootcamp 2026: Tutorial Proposal

Three-Tiered Learning Path

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San Gwann, February 2026

Overview: Target Audience & Structure

The tutorial is split into three distinct sessions to cater to different expertise levels:

- **Session #1 (Beginner):** Focus on Tabular RL and discrete MDPs.
- **Session #2 (Intermediate):** Continuous control and the "Reality Gap"
- **Session #3 (Advanced):** Custom environment design and competitive optimization.

Session #1: Foundations (Beginner)

Topic: Nascence of Q-Learning & SARSA

- **Environment:** Colab-based Maze environment (Discrete state-action space).
- **Key Concepts:**
 - ▶ Exploration vs. Exploitation.
 - ▶ Managing and updating Q-tables via sampling.
 - ▶ Introduction to the Bellman Equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

- **Goal:** Students understand how agents navigate grid-worlds without deep learning.

Session #2: Continuous Control (Intermediate)

Topic: Deep RL Robustness and Domain Randomization

- **Platform:** MuJoCo (Ant, Swimmer, Hopper).
- **Zero-Shot Transfer:** Testing "Healthy" Ant policies on "Crippled" environments.
- **Training Tasks:**
 - ▶ Evaluating performance drops (Percentage drop vs. Baseline).
 - ▶ Training specialist agents for "Hard" injury settings.
 - ▶ Hyperparameter tuning (Grid search on Learning Rate and Batch Size).
- **Toolkit:** Stable Baselines3 (PPO/SAC).

Session #3: The Flight Challenge (Advanced)

Topic: End-to-End MDP Design & Competition

- **Environment:** UMFlightEnv (Straight-line collision simulator).
- **Participant Responsibility:**
 - ▶ Design the **Observation Space** (5-tuple history) and **Action Space** (Clearances).
 - ▶ Craft the **Reward Method** (Balancing safety vs. efficiency).
- **Kaggle Leaderboard Objectives:**
 - 1 Primary: Zero collisions.
 - 2 Secondary: Minimal actions (clearances) used.
 - 3 Tertiary: Minimal unique aircraft involvement.
- **Draft Framework:** SB3 vs. Ray RLLib (TBD).

- **Workspace:** Proposed that students use Colab for $O(1)$ setup complexity
- **Compute:** All sessions optimized for CPU-only execution to avoid GPU bottlenecks with RL.
- **Configuration:** Still using Hydra for YAML-based experiment management (Session #2). Unclear whether this would be necessary for Session #3.
- **Evaluation:** Post-training analysis scripts will be provided for rendering MP4 videos of learned behaviours & extracting relevant metrics.